

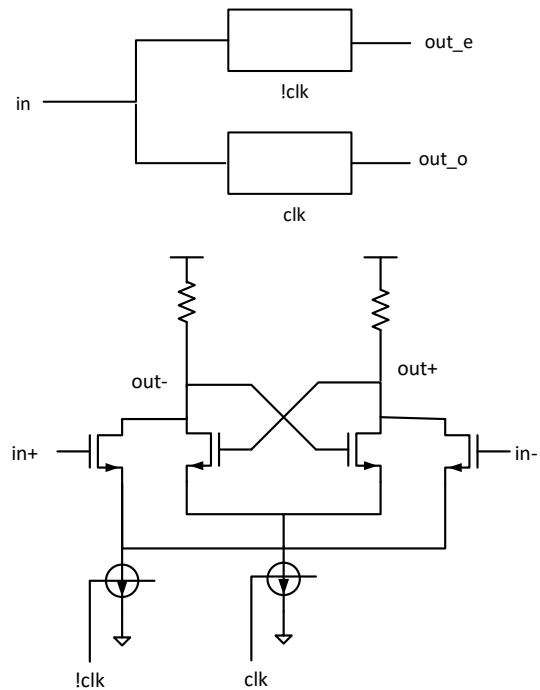
EE273 Digital Systems Engineering Problem Set #8

Assigned: May 28, 2026

Due: June 3, 2026

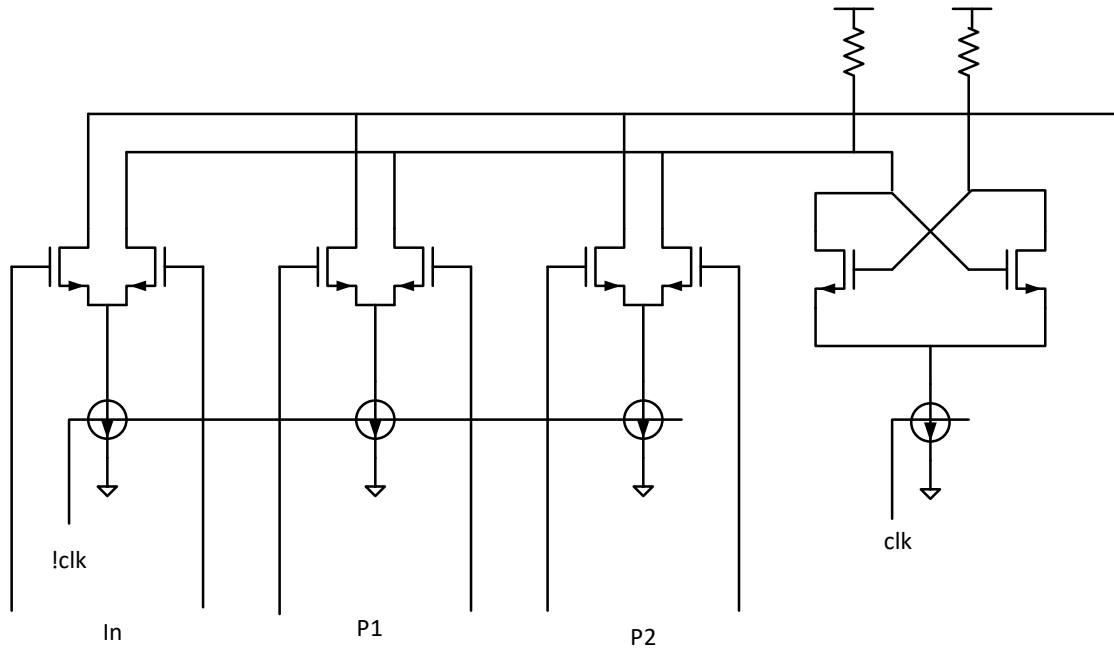
Last week you designed a CTLE to equalize a link, and this assignment is going to walk you through an analog DFE design for a 16Gbps link ($f_{nyq} = 8\text{GHz}$). The overall receiver is called a half-rate receiver, since the receive clock is at 8GHz, and the data is sampled at the rising and falling edge of the clock. Since the clock is half the data rate, you will need to build 2 input receivers on each pin. One receiver samples the even bits, and the other receiver samples the odd bits. The overall architecture that we are going to use was shown in Lecture 16 and is shown on the right.

This is the standard half-rate architecture which we need to extend for DFE. Normally each box would be a comparator. For high-speed circuits, these are generally built using MOS transistor to steer current, which is sometimes called current mode logic (CML). An outline of a normal comparator is shown on the right. There are two switched current sources on the bottom of the circuit which are controlled by the clock. In the sizing of this circuit, it is critical that there is gain from the input to the output. One also should be careful to keep the output swing small enough so that the transistors stay in saturation.

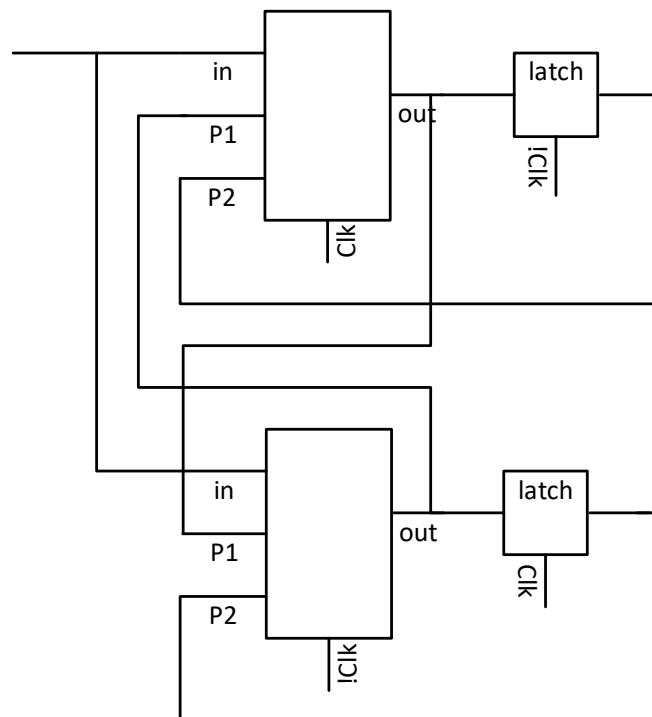


To make the DFE, we need to add a summer circuit between the CTLE output and the input of the latches. In a half rate architecture, it is better to add a summer to each receiver. In class I mentioned that you could include the summer inside of the latch to save a gate delay. This might work, but makes the simulation output more “interesting” to interpret. The current test bench has separated out the summing node from the latch.

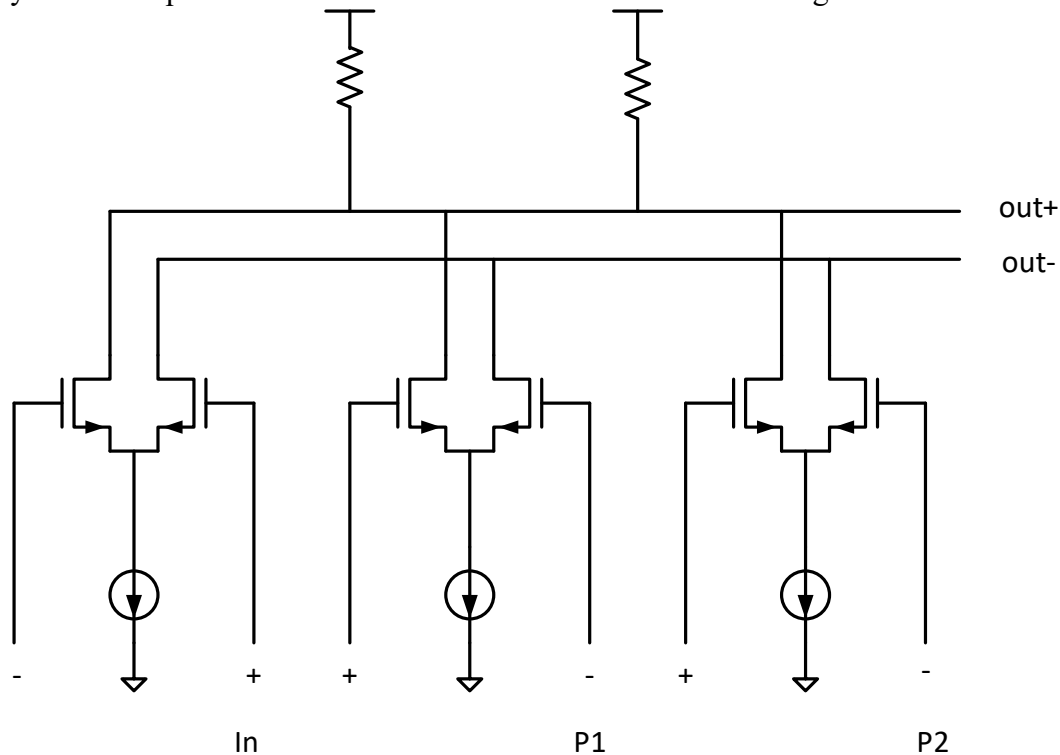
If you combine the summer and the latch you get a circuit that looks like:



And these would be wired together like:



If you use a separate summer circuit it will look like the following:



In this picture I put the sign of the inputs into the summer. For DFE, we want to subtract the post-cursor contribution from the incoming signal, so if the prior bit was 1, we would like to subtract its coefficient from the incoming signal. The current on each of the differential pair should be sized so the total current is constant (to keep the output swing constant), and the ratio of the currents should be set to the ratio of the cursor to the two post-cursor signals.

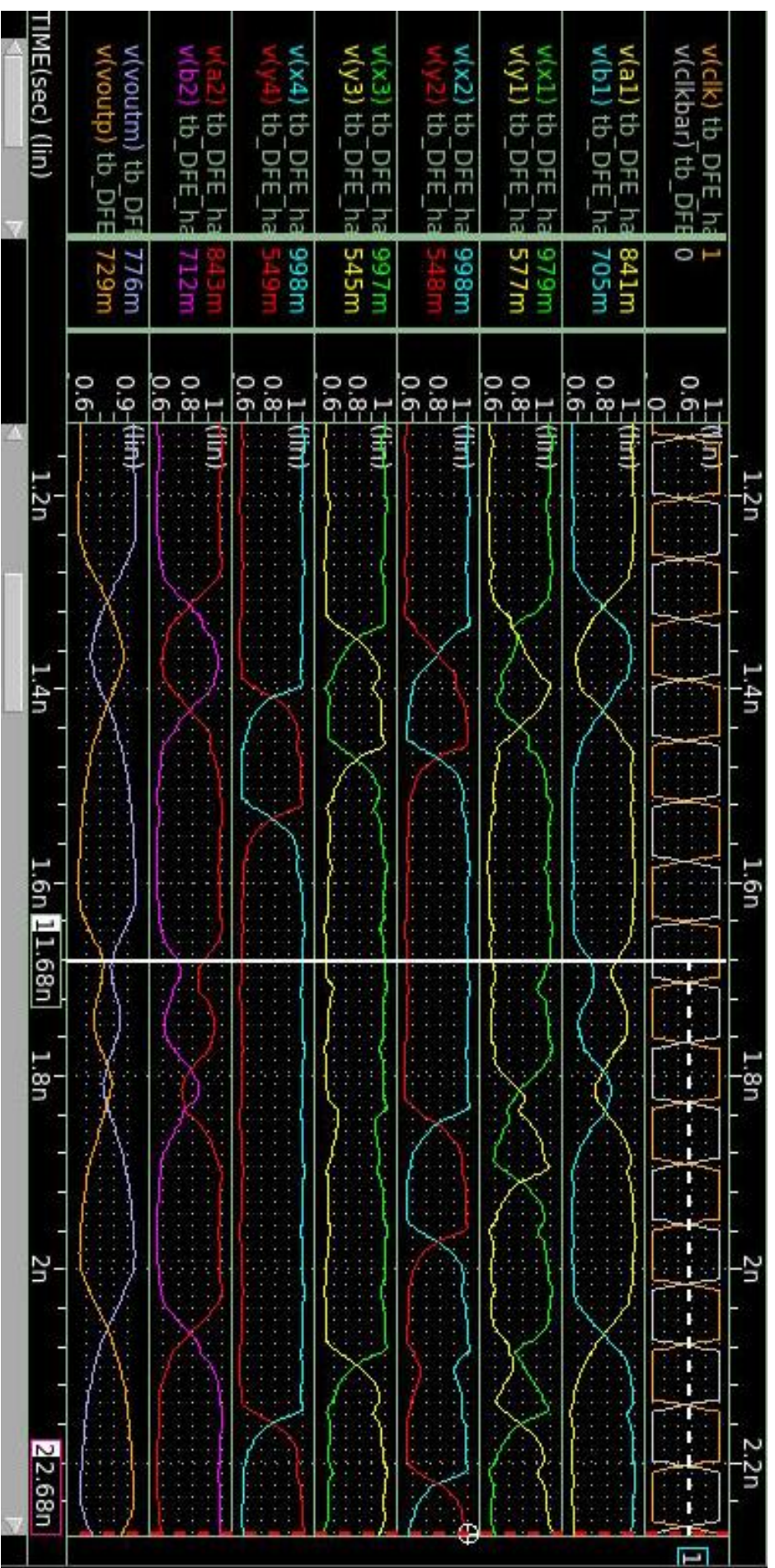
This problem is loosely based on Razavi's "Design of an Equalizer part II", which is worth reading, but this is not necessary to do the assignment. The testbenches for each part of the problem are already made for you, you just need to find a few critical values. Note: one UI is $1/16E9 = 62.5\text{ps}$.

1. (5 pts) When using a DFE the function of the CTLE changes. Since the DFE should remove the ISI for the first and possibly second post cursor, the CTLE should remove the precursor, and the later post cursors. We will work through this using the pulse_response testbench (in the tb directory). You should initially run the simulation to find the initial single bit response. This testbench will compute either the SBR, or the frequency response of the channel. If you look at the SBR, you will see it has a very slow tail that isn't good for DFE. To eliminate this tail, we want to create a zero at this pole frequency. Find the value of this time constant (it should be on the order of 100s of ps), or equivalently the -3dB frequency of the channel, and then set the cap of the CTLE to start boosting at this frequency. Please report the cap value that you used and show the resulting SBR and frequency response after the CTLE.
2. (5 pts) With this CTLE measure the cursor, and 1st and 2nd post cursor values. Report the 1st and 2nd post cursor values as a percentage of the cursor. You should also measure the precursor and 3rd post cursor values to make sure that these are small, and report these as well. You should notice that with the current CTLE the pulses don't cross, which means that the eye will be closed in this link without the DFE.
3. (5 pts) In the class we talked about the importance of choosing the right "gain" for the CML gates, and that gain is $gm \cdot R_L$. Since R_L is $V_{swing}/2I_d$ (where the I_d is at the bias point where the tail current is split evenly), the gain is $\frac{V_{swing}}{2} \frac{gm}{I_d}$. We will build our gates to have a V_{swing} of 400mV (you can play with that value if you would like) so to get a gain of at least 2 for each gate we need to have a gm/I_d of at least 10. Create a simple spice file to find the right current density (μA tail current to μm of transistor width) to get this gain / gm/I_d . There are a few ways to do this. One method is to create a simple differential pair. Set one of the gates to 1V (V_{dd}) and the other gate to 20mV below V_{dd} , effectively providing a $\pm 10mV$ input. Then connect each drain to V_{dd} using a separate power supply so you can measure the drain currents. Connect both sources to the same PWL current source. The difference in drain currents is $gm \cdot 20mV$. If you take the difference in current, and divide by the sum of the currents (which is the tail current) and multiple by 100, you will get gm/I_d .¹ Find the tail current where gm/I_d is 10.² Once you find this value, create the resistors to give 400mV swing, and check that the gain of this gate is indeed 2. Then set this value as the tail current in the 'sys_params.spi' file.
4. (5 pts) In building the summer, we have an interesting dilemma. We want the P1 and P2 inputs to be digital, so having gain of 2 gates is good, but the gain of the input should really be 1 for two reasons. First, we computed the percentage of the post cursor from the input, and not the input after some gain. Second, if the gain was 2, the gate would not be linear over the full range, which would be bad. This means that we need to change the size of the differential pair in the summer to make the gm/I_d of these transistors to be 5. Find out this ratio, test it in a gate to make sure it works, and put it in the sys_params.spi file as well.

¹ Since the input difference is 20mV, you might think you should multiply by 50. But you are dividing by the tail current, which is twice the current in each transistor, so you need to multiply by 100.

² We ran our simulations with the gm/I_d around 10% higher, at 11, to give more margin. You can do the same, or run it at 10.

5. (5pts) Unlike the CTLE, the DFE is clocked. This means that in order for the equalizer to work properly, we need to sample the bits at the right time. In the setup and hold time testbench, plot the values of V(A) and V(B) on a single plot. This is the differential voltage at the summer, which is what we want to sample with our first latch. Below this plot, plot the CLK and CLKBAR signals. Notice that in this testbench the data is running slightly faster than the clock, so cycle after cycle the clock crossing gets further behind the data crossing. Now, plot x1 and y1 on a single plot below the other two, as well as x2 and y2 on another plot. These are the latch outputs. Notice the second latch stays constant for most of the duration, until it flips. This flip indicates that the first latch is no-longer able to sample the input before the transition. Examine what happens to the first latch at this point; it gets close to latching the value from, a and b, but doesn't quite make it! The difference between the crossing of a and b to the clock crossing time is roughly the setup time. Copy and paste the measurements of your setup and hold time here along with the measured values.
6. (5pts) Now you need to instantiate the latches and summers in the half-rate DFE testbench with the right connections. These will automatically use the parameters we define in the sys_params file. Follow the diagram above, and make sure to double check the input and output polarities in the headers to ensure the correct connections. Copy and paste a screenshot of this filled out section of the file here.
7. (5pts) You've almost implemented the DFE now. First run the half-rate DFE testbench as is. View the waveforms for the clock, summing nodes, latches, and ctle output (like the screenshot below). Remember how the CTLE single bit response didn't quite cross for the single bit response? Find an example where this happens in the PRBS waveform. Notice our summing nodes also don't cross when this happens (yet). First thing to do is adjust the clock; we want the peak of our bits to happen a setup time before our latch samples (the falling edge of clk for latch1). Use your setup time measurement to adjust the **Tdelay** parameter for the clock, shifting it so that we sample the summer at the optimal time. The last thing to do is put in your measured relative magnitude of the first and second postcursors into PC1 and PC2 in sys_params... and if you did everything correctly, your summing nodes should now cross during the bit we identified earlier, and the latches should now be latching the correct values. You just made a DFE! Paste a screenshot of this here, along with a screenshot of the eye diagram of vout_dfe_diff, measuring the eye height.



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